

Taxation and Household Decisions: An Intertemporal Analysis

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Research questions

- ▶ Income tax revenues are the main source of government revenue for most countries (50% in the U.S.)
- ▶ Countries differ in how they tax married individuals
 - ▶ Individual taxation (e.g., Sweden, Canada, and the UK)

$$T_{married}(y_1, y_2) = T_{single}(y_1) + T_{single}(y_2)$$

- ▶ Taxation based on pooled income (the U.S. and Germany)

$$T_{married}(y_1, y_2) = \tilde{T}_{married}(y_1 + y_2)$$

- ▶ More general alternatives are possible
- ▶ **Which family-taxation system maximizes welfare?**
- ▶ **How do family-taxation systems affect welfare inequality?**

This paper

1. Discuss labor supply incentives in the U.S. tax system
 - ▶ Estimate the impact of taxation on secondary earners' employment using the 2003 Bush tax cuts
2. Build, estimate, and validate a life-cycle model of marriage, divorce, and household decisions
3. Calculate optimal tax parameters under four different family-taxation systems

Preview of Results

1. The U.S. joint system is close to the optimal income-splitting system (the optimal system is a bit more progressive)
2. The optimal individual-taxation system produces higher utilitarian welfare than the income-splitting system, but women with low earnings potential lose
3. A more general joint system generates the highest welfare, but is difficult to implement
4. **Income-splitting + a flat deduction for secondary-earner income** is simple, outperforms the individual system, and spreads welfare gains more evenly

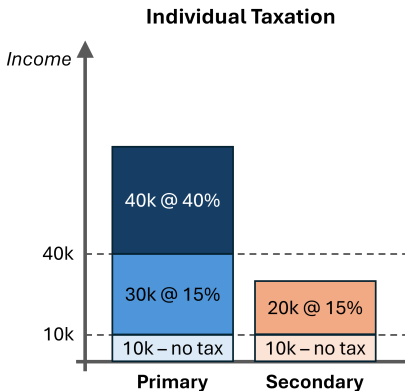
Related Literature

- ▶ Apps and Rees (1988, 1999a,b), Kleven, Kreiner, Saez (2009), Frankel (2014), Alves et al. (2023), Golosov and Krasikov (2023), Bierbrauer et al. (2023), ...
- ▶ Guner, Kaygusuz, and Ventura (2012), Gayle and Shephard (2019), Borella, De Nardi, and Yang (2022)
- ▶ Bick and Fuchs-Schuendeln (2017), Bick et al. (2019)

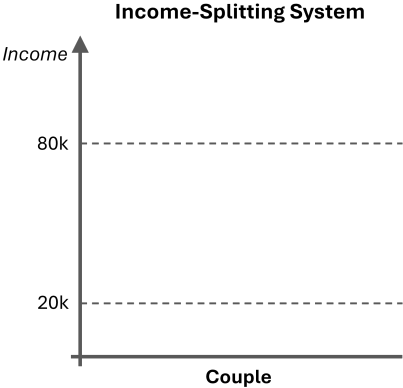
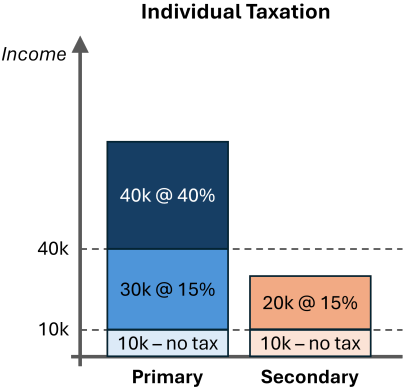
Outline

1. **Descriptive and Reduced-Form Evidence**
2. Model
3. Estimation and Validation
4. Comparison of Tax Systems
5. Conclusion

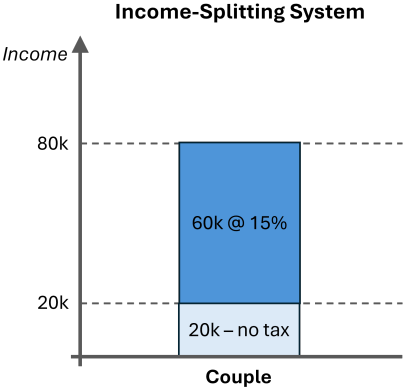
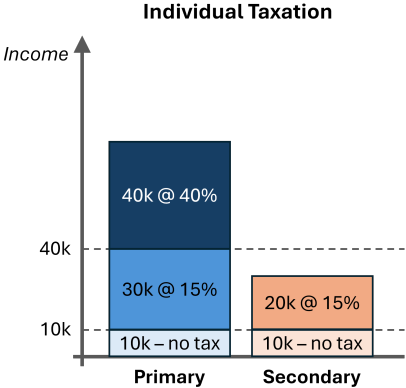
Income-Splitting vs. Individual Taxation: Example



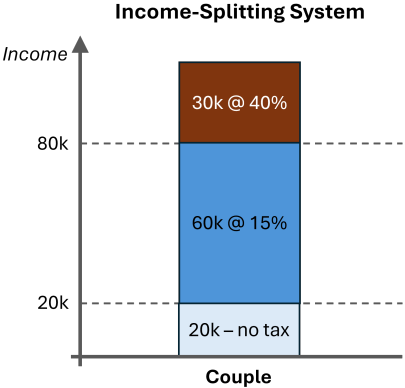
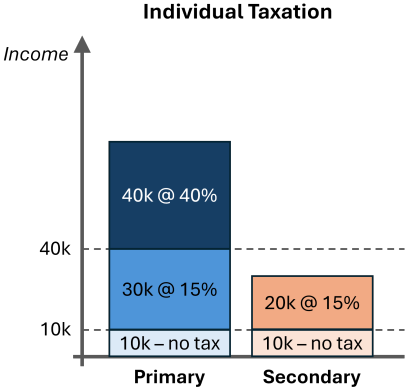
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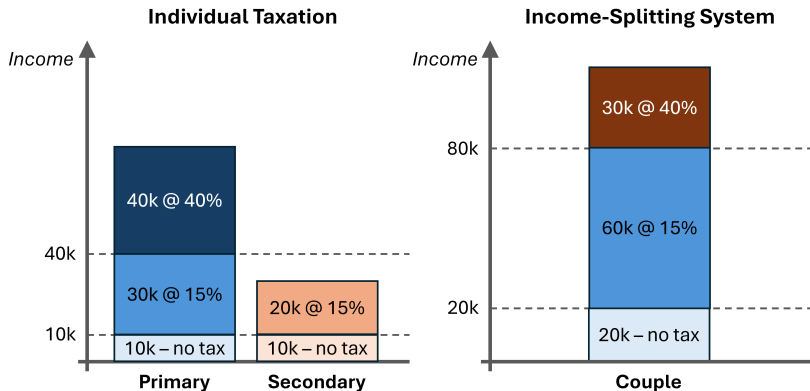
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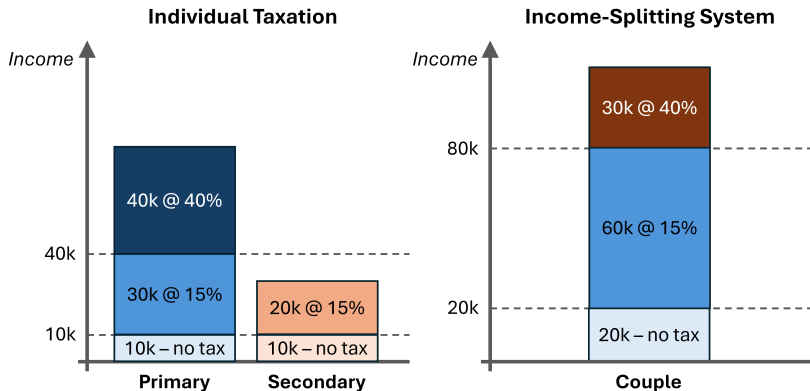
Income-Splitting vs. Individual Taxation: Example



Tax rates for the secondary earner:

- ▶ Marginal: 15% under Individual System, 40% under Income Splitting
- ▶ Average: 10% under Individual System, 40% under Income Splitting

Income-Splitting vs. Individual Taxation: Example

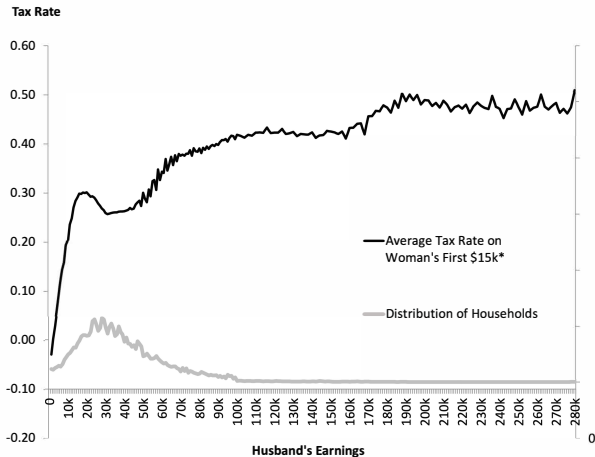


After-tax income for the couple:

- ▶ \$86.5k under Individual System, \$89k under Income Splitting
- ▶ The *marriage bonus* is \$2.5k
- ▶ Bonuses increase with specialization

Average Tax Rates

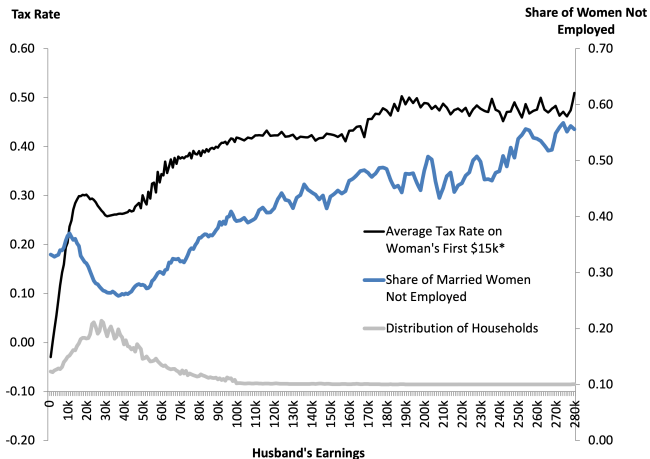
Married Women's Tax Rates, by Husband's Earnings



Source: NBER TAXSIM, ACS 2003-2006. Simulation for a household without dependents.

Average Tax Rates

Married Women's Tax Rates, by Husband's Earnings

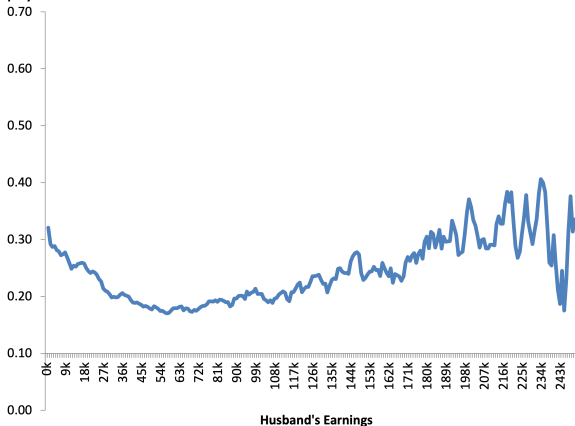


Source: NBER TAXSIM, ACS 2003-2006. Simulation for a household without dependents.

Average Tax Rates: Canada

Married Women Not Employed, by Husband's Earnings

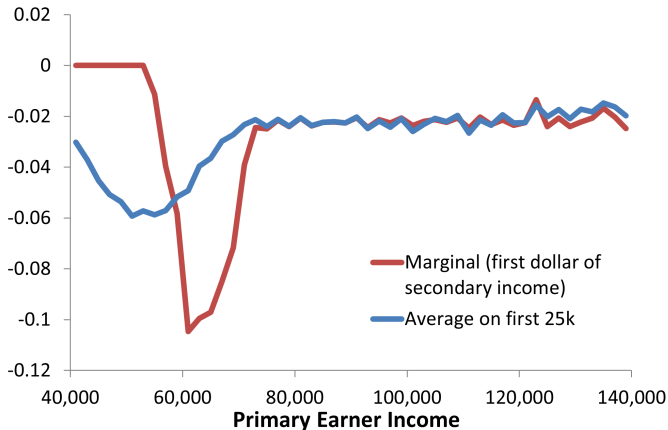
Share of Women Not
Employed



2003 Bush Tax Cuts

- ▶ Two main effects on personal income tax:
 - ▶ Top tax rates fall by about two percentage points
 - ▶ One threshold separating brackets moves down
- ▶ Consider families in which the **primary** earner has a good, full-time job
- ▶ Depending on how much the **primary** earns, taxes on *the secondary earner's income* may fall if the family is moved to a lower bracket
- ▶ Are secondary earners in those families more likely to work after the reform?

Tax Changes Induced by the Reform



- ▶ **Sample:** CPS, tax years 2001-2005; couples in which the primary earner works full-time, earns between \$40k and \$140k
- ▶ Tax rates are calculated using TAXSIM

Secondary Earner Responses to Bush Tax Cuts

$$SecondaryEarnerWorks_{it} = \beta \tilde{\tau}_{i,t} + \sum_{k=1}^{20} \alpha_k inc_{k,i,t} + \delta_t + \gamma X_{i,t} + \varepsilon_{it}$$

- ▶ $\tilde{\tau}_{i,t}$: Average tax on the first \$25k
- ▶ $inc_{k,i,t}$: The primary earner in family (i, t) is in income bin k

	(1)	(2)
	CPS Data	
$\tilde{\tau}_{i,t}$	-0.651*** (0.138)	-0.314** (0.144)
$\tilde{\tau}_{i,t} \times YoungChild_{i,t}$		-0.534*** (0.0762)
N	93115	93115

Why Use a Structural Model?

- ▶ Strong evidence that tax incentives matter
 - ▶ A shift from income-splitting joint taxation to individual taxation would increase secondary earners' labor supply
- ▶ But we also care about other outcomes, including welfare
- ▶ We also want to study alternative tax systems

Outline

1. Descriptive and Reduced-Form Evidence
2. **Model**
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Overview of the model

- ▶ Simulate the life cycle of a cohort of individuals
- ▶ The model begins at the start of working life; education and unobserved ability are exogenous
- ▶ T work periods, followed by R simplified retirement periods
- ▶ Choices in every work period:
 - ▶ Time split between labor, leisure, and household production
 - ▶ Resource allocation between consumption, savings, and market inputs used for household production
 - ▶ Marrying or not (if single); divorce or not (if married)

Single-period preferences

$$u(c^i, l^i, Q, \theta | \kappa, m) = \frac{(c^i)^{1-\sigma_c}}{1-\sigma_c} + \gamma_l \frac{(l^i)^{1-\sigma_l}}{1-\sigma_l} + \gamma_Q^{\kappa, m} \log Q + \theta$$

- ▶ c^i is private consumption, l^i is private leisure
- ▶ Q is the household *public good*
 - ▶ Produced using time and/or market-purchased inputs
 - ▶ The parameter $\gamma_Q^{\kappa, m}$ depends on child status (no children, children younger than 6, children older than 6) and marriage status
- ▶ θ is marriage quality ($\theta = 0$ for singles)

Work, wages, and human capital

- ▶ Four employment choices: none, part-time, full-time, more than full-time (45 hours)
- ▶ Pre-tax wage depends on education, gender, unobserved ability, and *accumulated human capital*
 - ▶ Estimated separately using a selection model
- ▶ Human capital accumulates when working and depreciates when not working

Marriage, divorce, fertility

- ▶ In every period, singles are paired with a potential match and get a random draw of marriage quality θ
 - ▶ Decide whether to marry based on lifetime expected utility
- ▶ Married households make choices efficiently but cannot commit to future allocations (Mazzocco 2007).
- ▶ Married individuals get marriage-quality shocks and may choose to divorce if the continuation value of marriage is too low.
- ▶ Childbirth is random for married women ages 20-35; children age stochastically
- ▶ Children affect production and the value of the public good Q ; young children also require daycare for working parents

Taxes, transfers, benefits

- ▶ We model in detail the following elements of the tax system:
 - ▶ Federal income tax schedule
 - ▶ Average state taxes
 - ▶ Earned Income Tax Credit
 - ▶ Child Tax Credit
 - ▶ Child and Dependent Care Credit
 - ▶ SNAP
 - ▶ SS/Medicare taxes

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Estimation

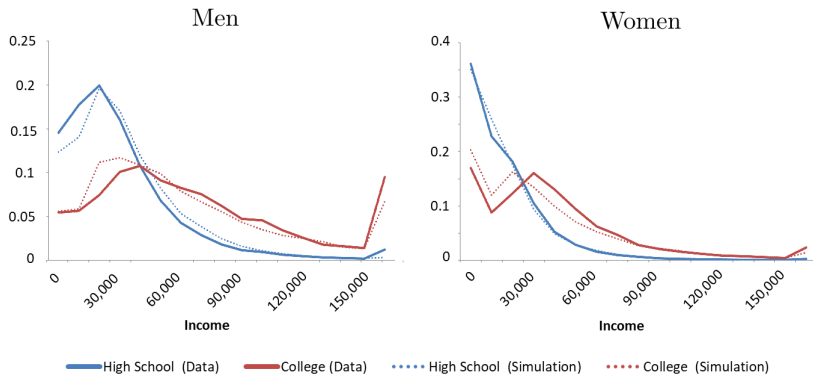
- ▶ Data sources: PSID, CPS, the American Time Use Survey (ATUS), and the Consumer Expenditure Survey (CEX)
- ▶ Some parameters are calibrated: $\beta = 0.98$, $\sigma_c = 1.5$
- ▶ Wage parameters and childbirth/child-aging processes are estimated externally
- ▶ All other parameters are estimated using the Simulated Method of Moments

Quality of fit, targeted moments (1/5)

Moment:	Data	Model
Share Married, Ages 22 to 60	0.61	0.65
Share Divorced, Ages 22 to 60	0.12	0.11
Share with a Child < 6, Ages 22 to 38	0.34	0.38
Weekly Hours Spent in Home Production:		
Men	12.3	14.1
Women	25.2	21.2
Sh. Households in Which the Woman Is the Higher Earner:		
When Husband Is a High School Graduate	29.3	33.7
When Husband Is a College Graduate	26.2	31.2

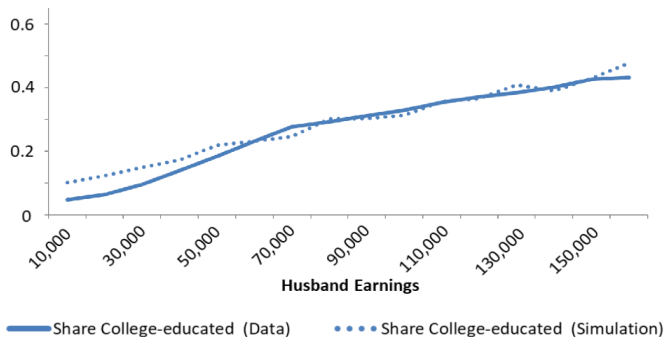
Quality of fit, targeted moments (2/5)

Income Distribution by Education (Ages 25-54)



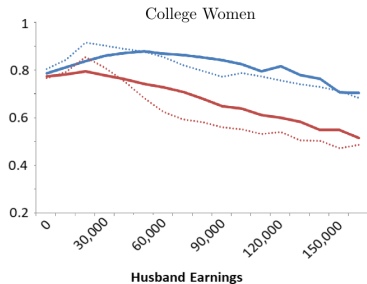
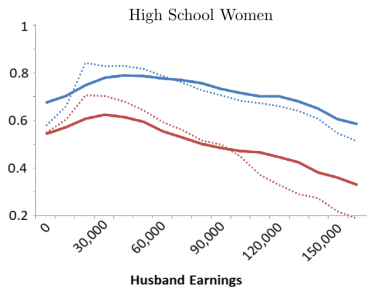
Quality of fit, targeted moments (3/5)

Share of Married Women with College Education, by Husband's Earnings



Quality of fit, targeted moments (4/5)

Share of Married Women Employed, by Husband's Earnings and Child Status



— No young children (Data) — Children < 6 (Data) No young children (Simulation) Children < 6 (Simulation)

Quality of fit, targeted moments (5/5)

Weekly Time Spent on Leisure: Data and Model Simulation

	No Children		Children 6+	
	Data	Model	Data	Model
All	66.1	66.4	59.3	59.9
Men	65.9	65.9	58.9	58.6
Women	66.4	66.9	59.8	61.2
High School	66.9	67.3	60.2	60.9
College	64.4	64.4	57.3	57.4
Married Women, by Quartile of Husband's Income				
First Quartile	64.5	65.3	59.9	60.5
Second Quartile	63.8	64.6	60.7	62.1
Third Quartile	65.0	66.3	59.0	60.4
Fourth Quartile	66.0	67.2	61.8	62.3

Validation with untargeted moments

- ▶ Frisch elasticities implied by the model are in line with the best available evidence
 - ▶ Interesting heterogeneity by gender, education, and family status [▶ Go](#)
- ▶ Effects of the Bush tax cuts:

	(1)	(2)	(3)	(4)
	CPS Data		Simulated Data	
$\tilde{\tau}_{i,t}$	-1.281*** (0.259)	-0.865*** (0.274)	-2.037*** (0.119)	-1.190*** (0.122)
$\tilde{\tau}_{i,t} \times YoungChild_{i,t}$		-0.617*** (0.142)		-1.282*** (0.0550)
N	25937	25937	123594	123594

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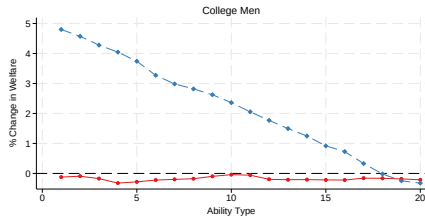
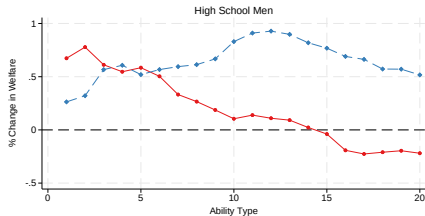
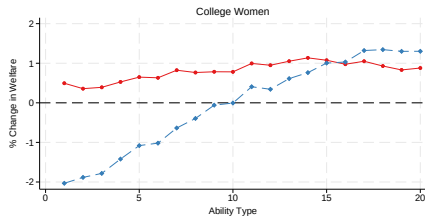
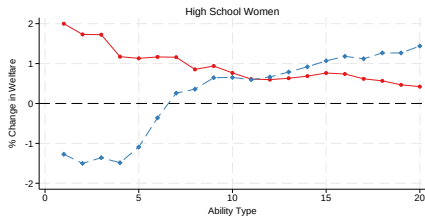
Policy Evaluation

- ▶ Using the estimated model, we evaluate four different income-tax systems
- ▶ For each system, we derive the tax rates that maximize total welfare in the economy, while holding constant:
 - ▶ Tax revenues
 - ▶ Tax brackets
 - ▶ The welfare system

Income-splitting vs. Individual Taxation

- ▶ How much are individuals willing to pay, on average, to move from the current system to alternative optimal systems?
 - ▶ Optimal joint with income-splitting: 0.9% of income
 - ▶ Optimal individual taxation: 1.25% of income
- ▶ The individual system raises women's labor force participation by about 10 percentage points [▶ Figure](#)
- ▶ Does this mean the individual system is the clear winner?

Income-splitting vs. Individual Taxation: Welfare Results



Income Splitting Individual

Income-splitting vs. Individual Taxation

What trade-offs constrain the policymaker?

- ▶ **Income-splitting**

- ▶ Hard to redistribute income—a high marginal tax on a high-earning individual also taxes their spouse!

- ▶ **Individual taxation**

- ▶ More progressive, but low-ability secondary earners do not benefit that much from lower taxes on their income
- ▶ Low-ability secondary earners lose substantially from the removal of marriage bonuses
- ▶ Women work substantially more, but consumption of the household public good falls substantially

Alternative systems

▶ **General joint system**

- ▶ In the style of Gayle and Shephard (2019)
- ▶ Gives the policymaker more flexibility, yielding much higher average willingness to pay: **2.59%**
- ▶ Very complex: 41 tax parameters! ▶ Optimal parameters

Alternative systems

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- ▶ Gives the policymaker more flexibility, yielding much higher average willingness to pay: **2.59%**
- ▶ Very complex: 41 tax parameters! ▶ Optimal parameters

▶ **Income-splitting with a secondary-earner deduction:**

- ▶ Optimal deduction: \$9,134
- ▶ Average willingness to pay: **1.3%**, with gains spread very evenly
- ▶ Women work more, but also purchase more market goods to maintain the level of household consumption

More in the paper

- ▶ How each tax system affects:
 - ▶ Time allocations: labor, leisure, household production
 - ▶ Leisure
 - ▶ Household Prod Time
 - ▶ Production of the household good
 - ▶ Consumption of Q
 - ▶ Human capital accumulation
 - ▶ HC
 - ▶ Marriage, divorce, and assortative mating
 - ▶ The joint system with a secondary-earner deduction leads to more stable marriages
 - ▶ Table
 - ▶ Sorting
 - ▶ Consumption risk
 - ▶ Go
- ▶ Additional exercise: how does limited commitment affect optimal tax rates?
 - ▶ Results

Outline

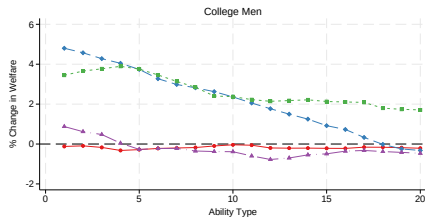
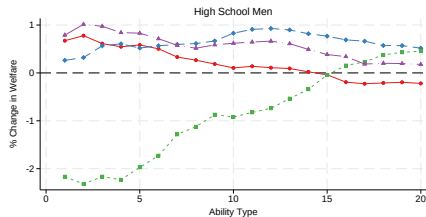
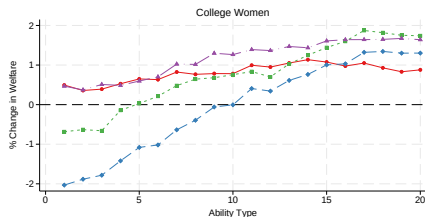
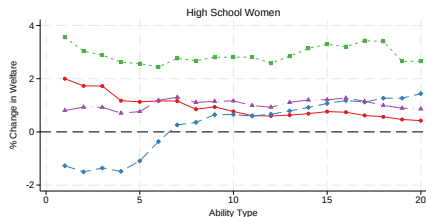
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Conclusion

- ▶ We document that secondary earners' employment choices respond to tax rates based on the primary earner's income level
 - ▶ Stronger responses in households with young children
- ▶ We build, estimate, and validate a life-cycle model that rationalizes this pattern and many others in the data
- ▶ We use the model to evaluate four potential policy reforms
 - ▶ Individual taxation: higher welfare than the income-splitting system, but also more inequality
 - ▶ An income-splitting system with a secondary-earner deduction achieves higher welfare, reduces welfare inequality, and is simple to implement

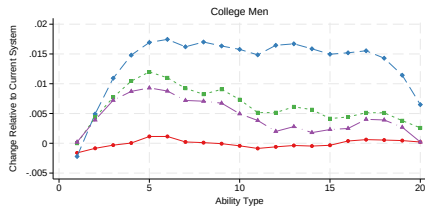
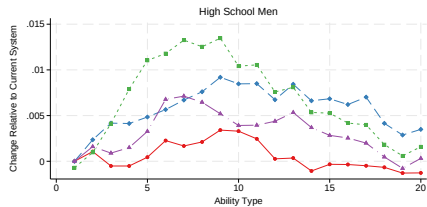
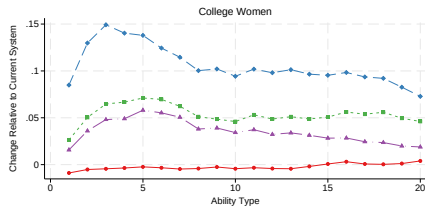
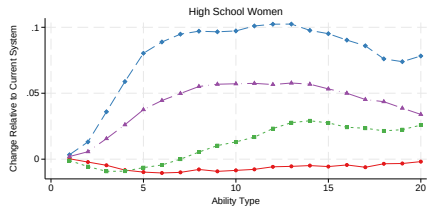
Thanks!

Welfare (willingness to pay for the reform)



Income Splitting Individual General Joint IS + SE Deduction

Labor Force Participation



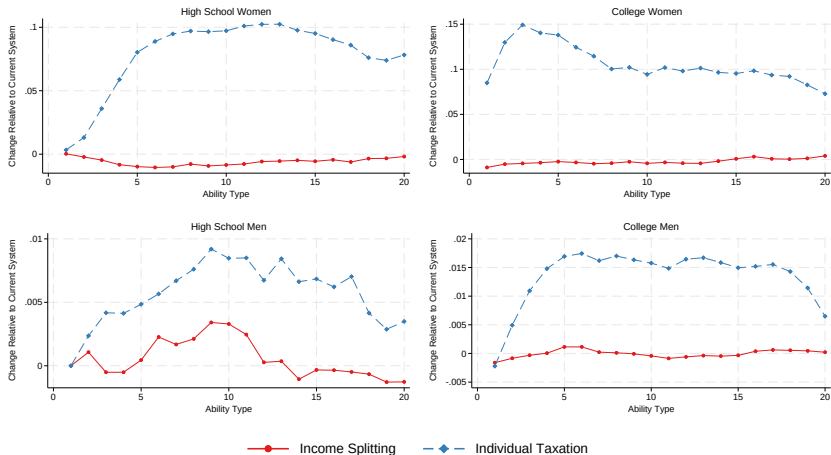
—●— Income Splitting
 —◆— Individual Taxation
 —■— General Joint
 —▲— IS + SE Deduction

Participation responses to temporary wage shocks [◀ Back](#)

	All	Single	Married	
			No yng. ch.	With yng. ch.
<i>Panel A: All women</i>				
Log wage temporary shock	0.580	0.585	0.562	0.633
Constant	0.712	0.718	0.736	0.590
<i>Panel B: Women, less than college</i>				
Log wage temporary shock	0.671	0.693	0.630	0.723
Constant	0.683	0.673	0.727	0.553
<i>Panel C: Women, college</i>				
Log wage temporary shock	0.421	0.321	0.468	0.507
Constant	0.780	0.864	0.753	0.658
<i>Panel D: All men</i>				
Log wage temporary shock	0.301	0.416	0.191	0.212
Constant	0.879	0.821	0.938	0.909
<i>Panel E: Men, less than college</i>				
Log wage temporary shock	0.351	0.484	0.194	0.282
Constant	0.856	0.792	0.934	0.882
<i>Panel F: Men, college</i>				
Log wage temporary shock	0.211	0.267	0.186	0.0969
Constant	0.930	0.904	0.945	0.968

Labor force participation responses

Change in Labor Force Participation

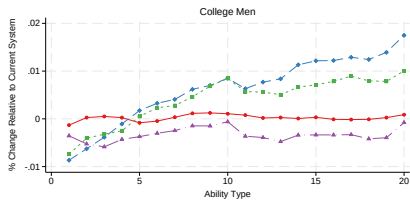
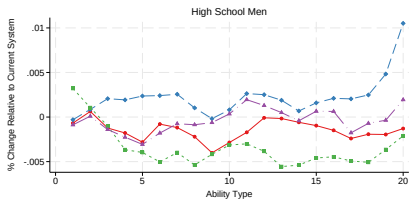
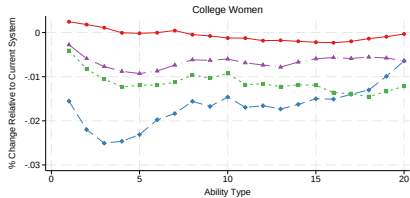
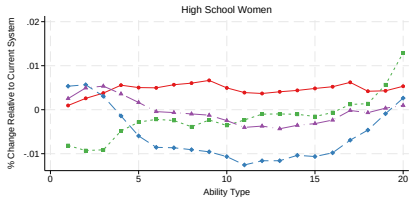


Policy Results: Optimal General Joint System

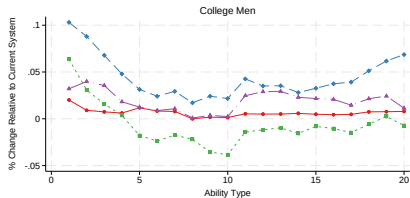
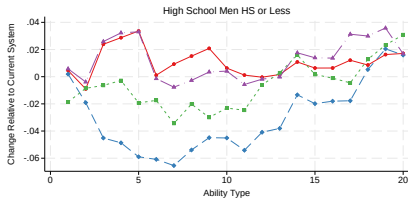
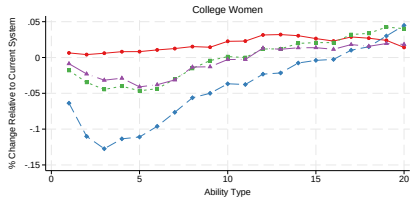
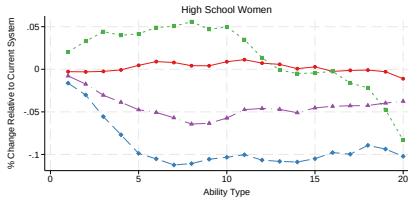
- Higher levels of redistribution without adversely affecting low-ability women

Income range							
	0-4.5	4.5-10.8	10.8-31	31-57	57-85	85-148	148-296
Marginal tax rates for single individuals							
	0.0%	33.1%	33.4%	46.9%	55.9%	59.6%	65.2%
Marginal tax rates for married individuals, by income of the spouse							
0	0.0%	0.0%	5.6%	8.5%	27.4%	38.1%	36.0%
4.5	11.0%	0.0%	7.7%	8.8%	28.4%	38.1%	36.2%
10.8	11.0%	9.4%	8.3%	11.0%	30.8%	37.8%	36.1%
31	20.6%	11.3%	31.0%	7.4%	36.4%	33.8%	37.7%
57	22.4%	20.2%	26.4%	17.2%	36.4%	38.9%	39.4%
85	28.6%	31.2%	34.2%	17.2%	31.6%	40.5%	38.4%
148	28.8%	28.5%	21.6%	29.4%	35.1%	33.1%	36.0%
296	34.3%	27.3%	33.4%	38.5%	30.0%	27.5%	23.6%
Income equivalent							2.59%

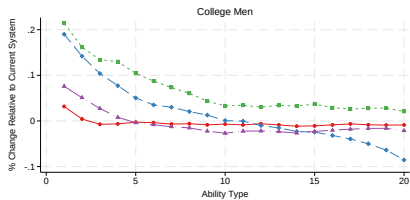
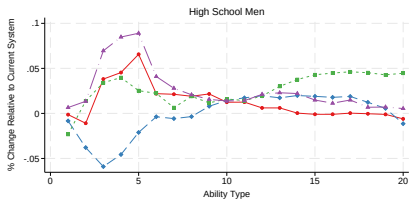
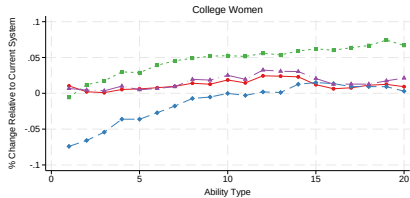
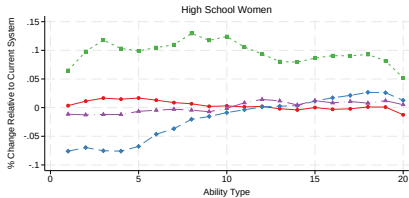
Golosov and Krasikov (2023): *Under realistic assumptions, optimal tax rates for married individuals (...) remain lower than the tax rates for single individuals, and the marginal rates for one spouse increase (decrease) in the earnings of the other if both spouses have low (high) earnings.*



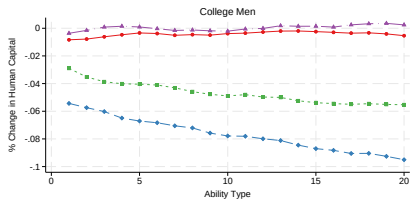
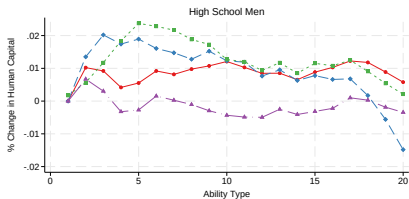
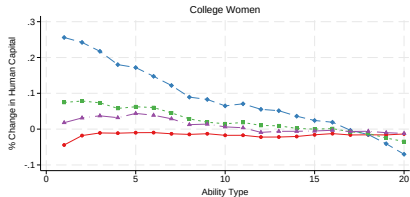
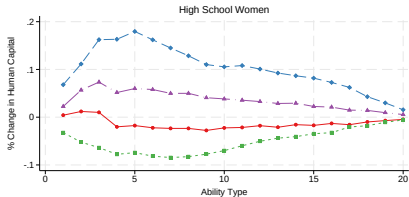
—●— Income Splitting
 —◆— Individual Taxation
 - - -■- - - General Joint
 —▲— IS + SE Deduction



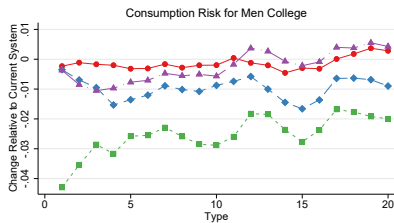
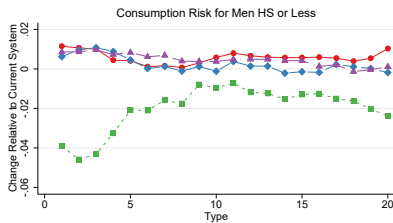
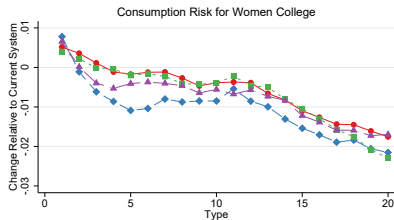
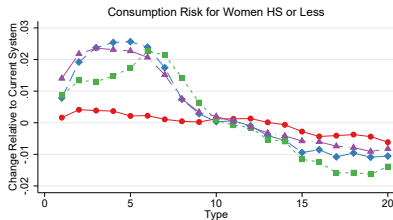
—●— Income Splitting
 —◆— Individual Taxation
 - -■- - General Joint
 —▲— IS + SE Deduction



—●— Income Splitting
 —◆— Individual Taxation
 - -■- - General Joint
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—●— Income Splitting
 —◆— Individual Taxation
 - -■- - General Joint
 —▲— IS + SE Deduction

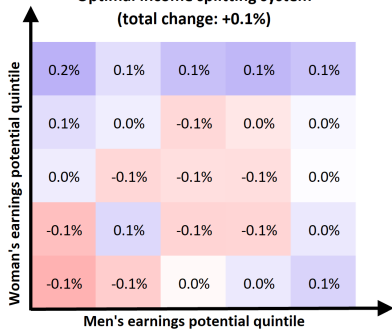


Notes: Moving averages.

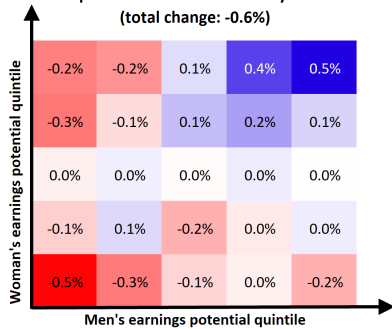
Tax Reform Effects on Marriage and Divorce

	Current	Splitting*	Ind.*	General*	Deduction*
<i>Panel A: All individuals</i>					
Share of adult life married	0.591	0.592	0.588	0.615	0.592
Share ever married	0.820	0.824	0.816	0.819	0.821
Share ever divorced	0.435	0.428	0.429	0.433	0.422
Corr. in earnings potentials	0.50	0.48	0.52	0.47	0.48
<i>Panel B: High school</i>					
Share of adult life married	0.565	0.564	0.554	0.583	0.559
Share ever married	0.815	0.821	0.814	0.815	0.820
Share ever divorced	0.454	0.444	0.445	0.453	0.437
Duration of first marriage	9.6	9.6	9.4	9.7	9.6
<i>Panel C: College</i>					
Share of adult life married	0.653	0.659	0.668	0.690	0.667
Share ever married	0.832	0.831	0.820	0.828	0.823
Share ever divorced	0.392	0.390	0.393	0.388	0.388
Duration of first marriage	11.7	11.8	11.9	12.2	11.9

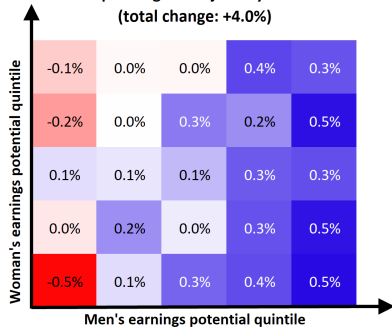
Optimal income splitting system
(total change: +0.1%)



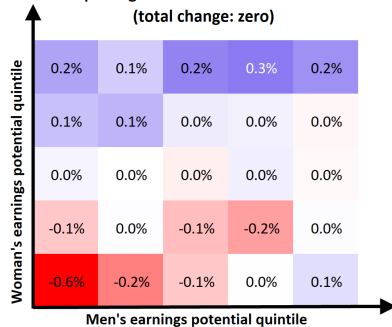
Optimal individual taxation system
(total change: -0.6%)



Optimal general joint system
(total change: +4.0%)



Inc. splitting with deduction for sec. earner
(total change: zero)



What is the effect of dynamics and limited commitment?

- ▶ We reduce the standard deviation of match quality shocks by 50%
- ▶ This decreases the average probability of divorce by 24%
- ▶ We then re-derive the optimal rates and deduction for the income-splitting system with a deduction

What is the effect of dynamics and limited commitment?

- ▶ The main effect is an increase in the degree of progressivity
- ▶ This is due to an increase in intra-household specialization

Bracket	Joint	Deduction	Deduction $0.5\sigma_\theta$
0-4.5k	0.0	0.0	0.0
4.5-10.75k	0.11	0.0	0.0
10.75-31k	0.19	0.25	0.24
31-57.25k	0.30	0.32	0.34
57.25-85.5k	0.33	0.37	0.39
85.5-149k	0.38	0.38	0.40
>149k	0.40	0.50	0.52
Optimal Deduction		\$9,134	\$9,382

Estimation: Moments

Table: Parameters Estimated Using SMM

Parameter	Interpretation	Heuristic Identification Arguments	Key Moments
<i>I. Household Production Function</i>			
τ	Substitutability between time in home production (d_i) and purchased market inputs (m)	It is identified by the elasticity of the ratio of the two inputs with respect to changes in individual wages, for single individuals, which measures the willingness to substitute between them.	V-VII
ν	Substitutability between the primary and secondary earners' time (d_1 and d_2)	It is identified by the elasticity of the ratio of the two time inputs with respect to the relative wages of primary and secondary earners.	V-VI
$\eta^{g, \kappa, r}$	Time inputs' share in the production function, for households with education g , child status κ , and marriage status r	Given τ , a higher η increases the fraction of resources allocated to time inputs in home production, relative to market inputs. Thus, η can be identified for different groups by the amount of market inputs used in household production divided by the sum of market inputs and time inputs valued at the individuals' wage.	V-VII
α_1, α_2	Productivity of time spent in home production for women and men	We normalize one parameter ($\alpha_2 = 2 - \alpha_1$) and identify the other by the difference between single men and women in the ratio of time inputs to the sum of the value of market and time inputs.	V, VI

Estimation: Moments

Table: Parameters Estimated Using SMM

Parameter	Interpretation	Heuristic Identification Arguments	Key Moments
<i>II. Preferences</i>			
γ_I	Taste for leisure relative to the taste for private consumption.	It can be identified by the average ratio of private consumption to leisure.	IV, VIII, IX
σ_I	Power parameter on leisure. The Frisch leisure elasticity of single individuals is equal to $-\frac{1}{\gamma_I \sigma_I}$.	Given γ_I , it can be identified by changes in leisure of singles in response to expected or marginal changes in their wages, i.e. changes that keep the marginal utility of wealth constant.	I-III, VIII, IX
$\gamma_Q^{K,r}$	Taste for public consumption relative to private consumption, by child and marriage status.	It can be identified by the average ratio of private consumption (or leisure, given γ_I) to the value of inputs used in household production, by child and marriage status.	V-X

Estimation: Moments

Table: Parameters Estimated Using SMM

Parameter	Interpretation	Heuristic Identification Arguments	Key Moments
<i>III. Marriage, Divorce, and Meeting Parameters</i>			
$\mu_\theta, \sigma_\theta$	Mean and standard deviation of initial match quality draw.	We normalize σ_θ to 1. Then, as a higher μ_θ increases the probability that a couple marries, we identify μ_θ by marriage probabilities by education and age.	XI
σ_z, K_{div}^g	Standard deviation of match quality shocks and cost of divorce.	The divorce probability is an increasing (decreasing) function of σ_z (K_{div}^g). We normalize σ_z to 1, and identify K_{div}^g using divorce probabilities for different education-age groups.	XI
π	Probability that an individual with a given education draws a potential spouse with the same education.	Higher π values increase the probabilities of meeting and, thus, marrying. We therefore identify π using the shares of individuals with high school or college education that marry a person with the same education.	XII
ρ_{g, g_s}	Correlation between abilities of potential spouses who have education g and g_s .	Higher values increase the probability that an individual with high ability meets and, thus, marries a partner who also has high ability. The parameters can therefore be identified from the correlation in spousal wages, conditional on education and human capital.	XII

Estimation: Moments

	Moment Description	Number
I.	Labor force participation by gender, education, marital status, and family status (no children, youngest child age 6+, youngest child under 6)	24
II.	Labor force participation by gender, education, and age group (20-25, 26-54, 56-64)	12
III.	Hours worked by gender, education, marital status, and family status (no children, youngest child age 6+, youngest child under 6)	24
IV.	Married women's labor force participation by income quantile of husband, education, and family status	30
V.	Hours in home production by gender, education, marital status, and family status	20
VI.	Value of hours in home production by gender, education, marital status, and family status	20
VII.	Expenditures on market good inputs on home production by gender, education, marital status, and family status	20
VIII.	Time spent on leisure, by gender, education, and marital and family status	20
IX.	Ratio between leisure and private consumption of married women without children, by husband's income quartile	8
X.	Ratio between private consumption and expenditure on market good inputs, by marital and family status	6
XI.	Share of women married and divorced by education and age group (22-26, 27-36, 37-50, above 50); share of women ever married by 36, by education; share of women ever divorced by 50, by education	12
XII.	Share of women and men with a high school or college education married to individual with the same education; correlation between permanent wage components, conditional on educations of husband and wife	8